UNIVERSITY OF MANITOBA

Department of Statistics

338 Machray Hall
Winnipeg, Manitoba
Canada, R3T 2N2
Telephone (204) 474-6274
Fax (204) 474-7621
Saman_Muthukumarana@UManitoba.CA

April 18, 2017

Accreditation Committee, Statistical Society of Canada
219 - 1725 St. Laurent Blvd.
Ottawa, ON, K1G 3V4

Re: Application Package for the SSC Accreditation for A. Stat Educational requirements

Dear Committee Members:
Please accept our renewal application package for the SSC Accreditation for A. Stat Educational requirements. We hope that you will consider our application though there is a lapse in renewal from 2012.

The courses that we offer towards the A. Stat Educational requirements are based on the ones approved in 2007, with the following modifications.

- Mathematics module is updated with equivalent courses due to the courses and programs changes introduced by the Department of Mathematics.
- Statistics and Probability modules are updated with equivalent courses due to the courses and programs changes introduced by the Department of Statistics.
- Computer Skills module is updated with equivalent courses due to the courses and programs changes introduced by the Department of Computer Science and Statistics.

Suggested courses that may be used towards the A.Stat. designation are attached in the file "courses.pdf". As requested, we also attach a zip file containing the details of these courses, including the course outlines, assigments, mid-term and final exams if applicable. For your convinience, a table of content of the zip file is given at the end of this letter. Please let us know if more information is needed.

Sincerely,

Saman Muthukumarana
Associate Head (Undergraduate), Department of Statistics

Brad Johnson
Associate Head (Graduate), Department of Statistics

Liqun Wang
Head, Department of Statistics

Stefi Alison Baum
Dean, Faculty of Science

University of Manitoba
Department of Statistics

Suggested courses that may be used towards the A.Stat. designation

Module	Course number, name (All STAT courses listed are 3 credit hours)
Mathematics Module	
1. Calculus I	MATH 1230 Calculus 1 and MATH 1232 Calculus 2, or, MATH 1500 Introduction to Calculus and MATH 1700 Calculus 2
2. Calculus II	MATH 2150 Multivariable Calculus, or MATH 2720 Multivariable Calculus
3. Linear Algebra	MATH 2090 Linear Algebra 2
Statistics and Probability Modules	
4. Mathematical Statistics	STAT 3800 Mathematical Statistics
	STAT 3050 Introduction to Probability Theory and Its Applications
	STATS 4100 Statistical Inference 1
5. Linear Regression	STAT 3470 Statistical Methods for Research Workers 1
6. Design of Experiment	STAT 3480 Statistical Methods for Research Workers 2
7. Survey Sampling	STAT 4520 Sampling Techniques 1
8. Stat Elective 9. Stat Elective 10. Stat Elective	Select three from STAT 3400 Introduction to Probability 2 STAT 4530 Design of Experiments 1 STAT 4590 Design of Experiments 2 STAT 3170 Statistical Quality Control STAT 3380 An Introduction to Nonparametric Statistics STAT 3490 Time Series Analysis STAT 4200 Statistical Inference 2 STAT 4170 Lifetime Data Analysis STAT 4530 Sampling Techniques 2 STAT 4630 Stochastic Processes STAT 4690 Applied Multivariate Analysis STAT 4700 Statistical Consulting STAT 7060 Advanced Theory of Probability STAT 7080 Advanced Statistical Inference

	STAT 7140 Linear Models STAT 7200 Multivariate Analysis STAT 7240 Advanced Topics 1 (Lifetime Data Analysis) STAT 7250 Advanced Topics 2 (Stochastic Processes) STAT 7260 Time Series
Computer Skills	
11. Computer Skills 1	COMP 1010 Introductory Computer Science or COMP 1012 Computer Science for Scientists and Engineers
12. Computer Skills 2	STAT 2150 Statistics and Computing or STAT 4600 Computational Statistics
Communication Skills	
13. Communication Skills	STAT 4700 Statistical Consulting or STAT 7310 Research Tools for Statistics or any University of Manitoba course requiring significant writing component as designated by (W).
Substantive Area	
14. Course 1	The Honours Program in Statistics at the University of Manitoba requires 18 credit hours of courses in an area of application, such as Actuarial Mathematics, Biological Sciences or Economics (other than Mathematics or Computer Science).
15. Course 2	
16. Course 3	

Note that a minimum grade of B- is required, if a course is to be submitted for credit towards the A.Stat. educational requirement.

Supporting Documents

for

SSC Course Accreditation Application

Department of Statistics

University of Manitoba

April 18, 2017

UNIVERSITY
OF MANITOBA

Contents

1 STAT 3170 Statistical Quality Control **1**

STAT 3170 Course Outline (Winter 2017) 1

STAT 3170 Midterm Exam 1(Winter 2017) 6

STAT 3170 Midterm Exam 2 (Winter 2017) 17

STAT 3170 Final Exam (Winter 2017) 27

2 STAT 3480 Statistical Methods for Research Workers 2 **37**

STAT 3480 Course Outline (Winter 2014) 37

STAT 3480 Midterm Test 1 (Winter 2014) 39

STAT 3480 Midterm Test 2 (Winter 2014) 43

STAT 3480 Project (Winter 2014) 52

STAT 3480 Final Exam (Winter 2014) 53

3 STATS 4100 Statistical Inference 1 **72**

STAT 4100 Course Outline (Fall 2012) 72

STAT 4100 Assignment 0 (Fall 2012) 74

STAT 4100 Assignment 1 (Fall 2012) 76

STAT 4100 Assignment 2 (Fall 2012) 78

STAT 4100 Assignment 3 (Fall 2012) 81

STAT 4100 Assignment 4 (Fall 2012) 83

STAT 4100 Assignment 5 (Fall 2012) 86

STAT 4100 Midterm Test 1 (Fall 2012) 89

STAT 4100 Midterm Test 2 (Fall 2012) 97

STAT 4100 Final Exam (Fall 2012) 107

4 STAT 4200 Statistical Inference 2 **125**

STAT 4200 Course Outline (Winter 2013) 125

STAT 4200 Assignment 1 (Winter 2013) 128

STAT 4200 Assignment 2 (Winter 2013) 131

STAT 4200 Assignment 3 (Winter 2013) 134

STAT 4200 Assignment 4 (Winter 2013) 136

STAT 4200 Assignment 5 (Winter 2013) 138

STAT 4200 Midterm Test 1 (Winter 2013) 140

STAT 4200 Midterm Test 2 (Winter 2013) 148
STAT 4200 Final Exam (Winter 2013) 156

5 STAT 4520 Sampling Techniques 1 **172**
STAT 4520 Course Outline (Fall 2016) 172
STAT 4520 Midterm Test 1 (Fall 2016) 176
STAT 4520 Midterm Test 2 (Fall 2016) 177
STAT 4520 Final Exam (Fall 2016) 178

6 STAT 7060 Advanced Theory of Probability **181**
STAT 7060 Course Outline (Fall 2014) 181
STAT 7060 Midterm Test (Fall 2014) 182
STAT 7060 Final Exam (Fall 2014) 183

7 STAT 3800 Mathematical Statistics **184**
STAT 3800 Course Outline (Fall 2016) 184
STAT 3800 Midterm Exam 1(Fall 2016) 190
STAT 3800 Midterm Exam 2 (Fall 2016) 197
STAT 3800 Final Exam (Fall 2016) 204

8 STAT 3050 Introduction to Probability Theory and Its Applications **213**
STAT 3050 Course Outline (Winter 2014) 213
STAT 3050 Midterm Exam 1(Winter 2014) 215
STAT 3050 Midterm Exam 2 (Winter 2014) 219
STAT 3050 Final Exam (Winter 2014) 223

9 STAT 3400 Introduction to Probability 2 **231**
STAT 3400 Course Outline (Winter 2011) 231
STAT 3400 Midterm Exam 1(Winter 2011) 236
STAT 3400 Midterm Exam 2 (Winter 2011) 238
STAT 3400 Final Exam (Winter 2011) 240

10 STAT 3490 Time Series Analysis **250**
STAT 3490 Course Outline (Winter 2016) 250
STAT 3490 Midterm Exam 1(Winter 2016) 251
STAT 3490 Midterm Exam 2 (Winter 2016) 254

STAT 3490 Final Exam (Winter 2016) 257

11 STAT 4530 Design of Experiments 1 **263**

STAT 4530 Course Outline (Fall 2015) 263

STAT 4530 Assignment 1(Fall 2015) 265

STAT 4530 Assignment 2 (Fall 2015) 267

STAT 4530 Midterm Exam 1(Fall 2015) 269

STAT 4530 Midterm Exam 2 (Fall 2015) 271

STAT 4530 Final Exam (Fall 2015) 273

12 STAT 3470 Statistical Methods for Research Workers 1 **275**

STAT 3470 Course Outline (Fall 2016) 275

STAT 3470 Assignment 1 (Fall 2016) 279

STAT 3470 Assignment 2 (Fall 2016) 280

STAT 3470 Assignment 3 (Fall 2016) 281

STAT 3470 Assignment 4 (Fall 2016) 282

STAT 3470 Midterm Exam (Fall 2016) 284

STAT 3470 Final Exam (Fall 2016) 293

13 STAT 3380 An Introduction to Nonparametric Statistics **305**

STAT 3380 Course Outline (Winter 2016) 305

STAT 3380 Assignment 1 (Winter 2016) 308

STAT 3380 Assignment 2 (Winter 2016) 309

STAT 3380 Assignment 3 (Winter 2016) 311

STAT 3380 Assignment 4 (Winter 2016) 312

STAT 3380 Midterm Exam (Winter 2016) 313

STAT 3380 Final Exam (Winter 2016) 316

14 STAT 4590 Design of Experiments 2 **323**

STAT 4590 Course Outline (Winter 2016) 323

STAT 4590 Assignment 1(Winter 2016) 325

STAT 4590 Assignment 2 (Winter 2016) 327

STAT 4590 Midterm Exam 1(Winter 2016) 329

STAT 4590 Midterm Exam 2 (Winter 2016) 331

STAT 4590 Final Exam (Winter 2016) 333

15 STAT 4690 Applied Multivariate Analysis **336**

STAT 4690 Course Outline (Winter 2016) 336

STAT 4690 Assignment 1(Winter 2016) 338

STAT 4690 Assignment 2 (Winter 2016) 350

STAT 4690 Assignment 3 (Winter 2016) 361

STAT 4690 Midterm Exam (Winter 2016) 380

STAT 4690 Data Project (Winter 2016) 389

STAT 4590 Final Exam (Winter 2016) 391

16 STAT 7140 Linear Models **407**

STAT 7140 Course Outline (Fall 2016) 407

STAT 7140 Midterm Exam 1(Winter 2016) 413

STAT 7140 Midterm Exam 2 (Winter 2016) 420

STAT 7140 Final Exam 1 (Winter 2016) 422

STAT 7140 Final Exam 2 (Winter 2016) 431

17 COMP 1010 Introductory Computer Science **433**

COMP 1010 Course Outline (Winter 2017) 433

COMP 1010 Final Exam (Fall 2016) 436

18 MATH 1230 Calculus 1 **453**

MATH 1230 Course Outline (Fall 2016) 453

MATH 1230 Final Exam (Fall 2016) 458